ENVRSUST 9014

Environmental Data Analysis and Visualization

Table of Contents

Course information 3
Course description 4
Course Structure 5
  Course timetable 6
Course materials 8
  Software requirements 8
  Pre-class work 9
Methods of Evaluation 10
    Exams (40%) 10
    National Pollutant Release Inventory Assignment (20%) 12
    Presentation (5%) 13
    Independent Project StoryMap (30%) 11
    Presentation (5%) 12
Additional Statements 14
  Class norms 14
  Use of Generative Artificial Intelligence 14
  Statement on Gender-Based and Sexual Violence 14
  Academic Offences 14
  Plagiarism 15
  Health and Wellness Services 15
  Accommodation and Accessibility 15
  Religious Accommodation 16
  Accessible Education (AE) 16
  Accommodation Policies 16

Course information

Instructor: Dr. Paul Mensink, North Campus Building 208, paul.mensink@uwo.ca

Course site: OWL Brightspace: https://westernu.brightspace.com/d2l/home/198357

Delivery format: In person. Required course activities include hands-on technical work, field-based activities, project studios, presentations, and group-based applied work.

Enrollment restriction: Enrollment in this course is restricted to graduate students in the Master of Environment and Sustainability program

Course description

In Environmental Data Analysis and Data Visualization, we will focus on the sourcing, scrutiny, and critical evaluation of environmental data. A major emphasis will be placed on developing transferable and technical skills in data management, visualization, statistical analysis, and interpretation. By working with datasets connected to real-world environmental issues, students will build the skills needed to identify patterns, evaluate evidence, communicate findings, and develop a deeper understanding of critical environmental challenges.

This fast-paced course is designed to challenge your problem-solving and analytical skills through a mix of individual and group assessments. Before each class, you will complete short technical exercises designed to build your confidence with environmental datasets and analytical workflows. In class, we will focus on applying these skills to real-world environmental problems, with an emphasis on hands-on practice, data interpretation, and clear communication.

Creativity will be encouraged throughout the course, particularly through an independent assignment that will challenge students to use their acquired technical expertise to create a final StoryMap and presentation on a current environmental issue. Students will be expected to source, analyze, visualize, and interpret data in a way that is both technically sound and accessible to a broader audience.

Through our exploration of environmental data, we will cover a broad range of environmental topics, including biodiversity loss, invasive species, air and water pollution, land use change, climate change, and environmental monitoring. Across these topics, the focus will remain on how data can be used to understand complex environmental challenges, evaluate uncertainty, and communicate meaningful insights.

Learning outcomes

By the end of the course, students will be able to:

- Apply interdisciplinary data analysis approaches to investigate complex environmental problems by integrating field, statistical, geospatial, and open-source datasets to interpret patterns, evaluate evidence, and communicate findings.

- Develop problem-solving, decision-making, and critical analysis skills within the context of working with and presenting environmental data.

- Conduct environmental field data collection and integrate field observations with statistical, geospatial, and open-source datasets to support environmental assessment and decision-making.

Course Structure

This course is organized into two complementary units. In Unit 1, we focus on developing a broad suite of data analysis skills across multiple software platforms. Skills are introduced incrementally, emphasizing common analytical concepts and workflows across tools (e.g., joins in Excel and R alongside spatial joins in QGIS). This approach allows us to build technical proficiency efficiently while developing a versatile toolkit for data analysis. Unit 1 concludes with a series of examinations at the end of October to assess your core competencies and technical skills. In Unit 2, students apply their acquired skills through two major projects:

- **Independent Project StoryMap** – students conduct an extensive exploratory data analysis by sourcing openly available datasets, organizing them, and analyzing them to generate evidence-based insights into their focal environmental area.
- **National Pollutant Release Inventory (NPRI) Challenge** – our class will work in partnership with the NPRI team at Environment and Climate Change Canada to answer pressing questions about releases across the country.



Course timetable

Field work | Assessments | Project studio workshops

No.	Topic		Date
UNIT 1 – TECHNICAL SKILLS			
1	Topic 1: Loading, Cleaning and Exploring Data	Beneath the Surface: Measuring Animal Abundance in the Ocean	Sept 8th
2	Topic 2: Transforming and Wrangling Data	Animal Movement Across Seascapes: Tracking Marine Megafauna	September 14th
3	Topic 3: Conditional logic, conditional calculations, and deterministic modelling	Food for Thought: Fisheries data and management in Canada	September 15th
4	Topic 4: Field Data Collection	Biodiversity and Invasive Species Monitoring Field data collection (Assigned sections)	September 21st
5	Topic 5: Connecting Data - Joins, merges, and spatial relationships	Troubled Waters: The provincial water quality monitoring network	September 28th
6	Topic 6: Exploratory Data Analysis - Finding Patterns and Trends	Pollution Pathways: Tracking Canada's Environmental Releases	September 29th
7	Topic 7: Statistical Analysis - Testing Relationships and Interpreting Evidence	Forecasting Footprints: Canada's GHG Emissions and Climate Action	October 5th
8	Topic 8: Large-scale Geospatial Analysis - Scaling Patterns Across Landscapes	Landscapes in Motion: Measuring the Cost of Land Use Changes and Habitat Degradation	October 6th

9	Buckthorn Bust! Invasive Species Removal Evidence-based buckthorn intervention (All sections together)	Friday, October 9th
10	Core skills technical exam (Day 1) (All sections together)	October 26th 9–12
11	Core skills technical exam (Day 2) (All sections together)	October 27th 9–12
UNIT 2 – APPLIED PROJECT STUDIO		
12	**Data to Decisions: Communicating Insights from your Data** Visual storytelling, knowledge mobilization, and evidence-based communication	November 9th
13	**Independent Project Workshop** Students will refine their project scope, data sources, analytical approach, and visualizations. **Lightning talks:** Students will provide brief presentations of project concepts and progress updates.	November 10th
14	Individual presentations (Day 1) (Assigned sections)	November 23rd
15	Individual presentations (Day 2) (Assigned sections)	November 24th
16	NPRI Final Presentations NPRI groups will present to representatives from Environment and Climate Change Canada	December 2nd

Course materials

Software requirements

In this course, we will be using freely available and/or open-source software[1]. *Prior* to the beginning of the course, you will be required to download the following programs:



Microsoft Excel - All of you have access to Office 365 through your Western accounts and can download its desktop applications. You can log into your account with your Western credentials (https://portal.office.com/) and then click on "Install apps" in the top right corner to install Excel on your desktop.



R is a program for statistical computing and data visualization. You can download R here: https://cran.r-project.org/



Positron is an integrated development environment (IDE) that acts as a code editor for R. It makes working with R more intuitive and provides helpful functions for viewing data and previewing figures. Although you will download R, Positron is the main program we will use to access R. You can download Positron here: Download Positron – Positron



QGIS is an open-source geographic information system (GIS). This program is compatible with both Macs and PCs. You can download QGIS here: https://qgis.org/en/site/

Please be aware that if you are a Mac user, you will need to ensure that your operating system is updated to the latest version, and you may need to approve the app download in your security settings. The MacOS instructions are on the download page.

[1] Most of the software we use is open-source. I believe using open-source software is beneficial for you because institutional licenses for proprietary software often stop when you leave the university, which means you won't be able to access those programs after you graduate.

Pre-class work

Our class uses a partially flipped format. In a flipped classroom, you complete preparatory work before arriving, which helps make class time more engaging and effective. For each session, you will be expected to complete pre-class work before the lesson begins. Once you arrive to class, you will answer a short quiz on the pre-class work to assess your understanding and then we will use our in-person time to reinforce key concepts and focus on hands-on, practical application.

Before each class, you will complete work under the following categories:

- **Video tutorials** (found under Course topics and training): In each class, we will develop a set of practical technical skills used in environmental data analysis. Before arriving to class, you will work through a group of topic videos that guide you step by step through the tools and techniques we will use that day. In class, you will apply these skills to new datasets, using them to explore practical questions and draw insights from environmental data.

- **Readings** (located in pre-class work under each class): In each class, we will be working through a real dataset. To help you understand the dataset we will be using, you may be assigned a short selection of readings on that topic. These will include scientific articles, government reports, and news stories, providing background and real-world context for our in-class discussions. The readings are designed to help you engage with key issues and provide context for our in-class data analysis.

Methods of Evaluation

The course assessments are designed to develop your practical and cognitive skills. In addition to summative assessments, students will also be expected to make meaningful contributions to class discussions.

Category	Type	Deadline/Date	Weighting
Core Concepts and Technical Skills (40%)			
Exams	Individual	October 26th/27th	40%
Applied Projects (60%)			
Final presentation	Individual	November 23rd November 24th	5%
NPRI Presentation	Group	December 2nd	5%
Independent Project StoryMap	Individual	December 11th	30%
NPRI assignment report	Group	December 16th	20%

Attendance is a course requirement and failure to attend a sufficient number of classes may result in failure of the course.

Core Concepts and Technical Skills (40% of overall mark)

Exams (40%)

Open-Book Format

You will have two days of exams that will ask you to apply the knowledge and technical skills you have learned over Unit 1 of the course. The exams will assess the development of your technical skills (e.g., data manipulation and visualization). The exams will be cumulative and cover the entire range of the course material before the exam across Excel, QGIS and R. This open-book exam will assess your proficiency in Excel, R, and QGIS focusing on skills in data manipulation, analysis and visualization.

**Generative AI tools (e.g., ChatGPT, Copilot, Gemini) will be permitted for use in the all of the exams for the course. **

Applied Projects (60%)

Independent Project StoryMap (30%)

The major project for this course is an independent exploratory data analysis assignment where you will study a topic of your choice. This project is intentionally open-ended, offering you the flexibility to define a reasonable scope, source relevant datasets, and conduct meaningful analyses. It challenges you to thoughtfully plan and execute your data exploration from start to finish. A defining aspect of this project is the sourcing of data from open-source databases and repositories. You'll take the lead in combining and analyzing datasets to uncover meaningful relationships and answer the key questions you have identified for your project. You will select a broad environmental issue and evaluate it through the exploration of temporal, spatial, and other relevant environmental data.

The final deliverable for your project is a comprehensive ArcGIS storymap that highlights the results of your analysis and provides interpretation by effectively integrating text, figures, maps, and tables into one flowing document. We have scheduled an elective ArcGIS storymaps and dashboards workshop for you on November 6th and unless you feel very confident using storymaps and dashboards already, I would strongly recommend that you attend that workshop. Storymaps are a fun and interactive way to share data visualizations (we'll review examples throughout the course) and this project will give you (1) practice on how to best use storymaps and dashboards and (2) a final deliverable you can share with potential co-op employers.

For your storymap, you will be asked to include the following:

- **Infographic** - You will create a `social media ready` infographic that provides an overview and summary of your analysis for the public.
- **Figures** – You will need 10 figures created using R
- **Maps** - 10 maps created in QGIS (with at least two interactive maps)
- **Dashboard** - at least one dashboard
- **Written summary** - A written interpretation to accompany your data visualizations and analysis that summarizes your key findings. Make sure to cite all figures and maps that you include. When interpreting your results, incorporate references from academic literature to provide context; however, the emphasis should be primarily on your own analysis and interpretation of the open-source data you have gathered.
- **A list of data sources** - Provide a list of data sources you used for your project with weblinks to each of them.

- **References section** - include any academic references you have cited in your assignment. Use APA 7th edition for your reference format.

All figures, graphs, or maps should be created (by you) from data you have collected from various open sources. Please note, you will also submit your R scripts as part of your final submission.

Indepdent Project Presentation (5%)

On November 23rd or November 24th, you will present to the class covering your main findings. You will have an opportunity to take on feedback from this session before submitting your final StoryMap. Your presentation will be about 10 minutes, with time for questions afterwards. You will use a slide deck to take us through your work rather than presenting directly from your StoryMap, and you should ensure it includes the following:

- Title slide – A title for your project and your name
- Introduction slides (one or two) – Provide some basic background information about the topic.
    - What was the issue you explored?
    - Why did you explore that issue?
    - What is currently understood about it?
    - Where did you find the data for your project?
- Content slides – A range of slides showing some of your most compelling figures and maps. You should explain each of the figures and the interpretation within the context of your overall project. Please note, you do not have to share all your figures or maps and should focus on the figures that you find most impactful.

National Pollutant Release Inventory (NPRI) Assignment (20%)

We are collaborating with Environment and Climate Change Canada (ECCC) to answer pressing research questions about Canada's non-GHG emissions by using data from the National Pollutant Release Inventory. You will work in pre-assigned groups of five to six students to create a report for ECCC on various pre-defined topics.

You will submit a written report to your ECCC mentor for review that includes figures, maps and tables and a write-up on your findings. You can find more detailed guidance and past examples of reports on our NPRI Group Assignment page on OWL Brightspace.

Note: You will receive feedback from your ECCC mentor and your team will be responsible for responding to that feedback and submitting the final copy back to

ECCC. You will submit this finalized copy with track changes for a portion of your assignment grade.

NPRI Presentation (5%)

You will also give a virtual presentation to the ECCC mentors at the beginning of class on December 2nd. Each group will present detailed findings from their NPRI analysis, focusing on their assigned topic area and the key patterns, trends, or concerns identified through their data work. The presentation should translate your analysis into clear, evidence-based insights for the ECCC team and should highlight the figures, maps, tables, or other visualizations that best support your interpretation.

Where appropriate, groups should also identify potential recommendations that emerge from their findings. These may include suggestions for improved monitoring (e.g., changes in monitoring thresholds), areas where further investigation may be warranted, or priority areas that could help guide future analysis and decision-making. The goal is to present your work in a professional format that is useful to an applied audience and demonstrates how NPRI data can support environmental monitoring and policy-relevant decision-making.

Additional Statements

Class norms

Attendance at all sessions in this course is mandatory, and it is your responsibility to advise the instructor and seek assistance if you are unable to contribute to class discussions. Circumstances may arise that make it impossible for you to attend, including health-related reasons. If you are unable to meet a course requirement due to illness or other serious circumstances, please contact the instructor.

Under University regulations, I can determine at what point absenteeism has become excessive and thus may prevent you from passing the course. Any student who has not attended enough sessions to fully master the material may not be permitted to receive credit for the course.

Use of Generative Artificial Intelligence

Generative AI tools may be used in this course to support learning, coding, troubleshooting, data analysis, and communication. Students remain responsible for verifying the accuracy of all outputs and for ensuring that submitted work reflects their own understanding. Specific guidance regarding permitted and prohibited uses of AI will be provided for individual assessments.

Statement on Gender-Based and Sexual Violence

Western is committed to reducing incidents of gender-based and sexual violence (GBSV) and providing compassionate support to anyone who is going through or has gone through these traumatic events. If you are experiencing or have experienced GBSV (either recently or in the past), you will find information about support services for survivors, including emergency contacts on the Wellness and Wellbeing website. To connect with a case manager or set up an appointment, please contact support@uwo.ca.

Academic Offences

Scholastic offences are taken seriously and students are directed to read the Scholastic Offences policy, specifically, the definition of what constitutes a Scholastic Offence.

Scholastic offences are taken seriously, and students are expected to read the appropriate policies.

- The Code of Student Conduct: http://www.uwo.ca/univsec/pdf/board/code.pdf

Scholastic Discipline for Graduate Students: http://www.uwo.ca/univsec/pdf/academic_policies/appeals/scholastic_discipline_grad.pdf

- The Non-Discrimination/Harassment Policy: http://www.uwo.ca/univsec/mapp/section1/mapp135.pdf
- The Code of Behavior for Use of Computer Resources and Corporate Data: http://www.uwo.ca/univsec/mapp/section1/mapp113.pdf
- Graduate Student Academic Appeals: http://www.uwo.ca/univsec/pdf/academic_policies/appeals/appealsgrad.pdf

Plagiarism

Students must write their essays and assignments in their own words. Whenever students take an idea, or a passage from another author, they must acknowledge their debt both by using quotation marks where appropriate and by proper referencing such as footnotes or citations. Plagiarism is a major academic offence (see Scholastic Discipline for Graduate Students in the Western Academic Calendar).

All required papers may be subject to submission for textual similarity review to the commercial plagiarism detection software under license to the University for the detection of plagiarism. All papers submitted will be included as source documents in the reference database for the purpose of detecting plagiarism of papers subsequently submitted to the system. Use of the service is subject to the licensing agreement, currently between Western University and Turnitin.com (hyperlink www.turnitin.com).

Health and Wellness Services

As part of a successful graduate student experience at Western, we encourage students to make their health and wellness a priority. Western provides several on campus health-related services to help you achieve optimum health and engage in healthy living while pursuing your graduate degree. Students seeking help regarding mental health concerns are advised to speak to someone they feel comfortable confiding in, such as a faculty member, the Program Chair, or the MES Coordinator. Students who are in emotional/mental distress should refer to Mental Health@Western (https://uwo.ca/health/) for a complete list of options about how to obtain help.

Accommodation and Accessibility

Please contact the course instructor if you require lecture or printed material in an alternate format or if any other arrangements would make this course more accessible to you. You may also wish to contact Accessible Education at

http://academicsupport.uwo.ca/accessible_education/index.html if you have any questions regarding accommodations.

Religious Accommodation

When a course requirement conflicts with a religious holiday that requires an absence from the University or prohibits certain activities, students should request accommodation for their absence in writing at least two weeks prior to the holiday to the course instructor and/or the Academic Counselling office of their Faculty of Registration. Please consult University's list of recognized religious holidays (updated annually) at

https://multiculturalcalendar.com/ecal/index.php?s=c-univwo.

Accessible Education (AE)

Western is committed to achieving barrier-free accessibility for all its members, including graduate students. As part of this commitment, Western provides a variety of services devoted to promoting, advocating, and accommodating persons with disabilities in their respective graduate program. Graduate students with disabilities (for example, chronic illnesses, mental health conditions, mobility impairments) are strongly encouraged to register with Accessible Education (AE), a confidential service designed to support graduate and undergraduate students through their academic program. With the appropriate documentation, the student will work with both AE and their graduate programs (normally their Graduate Chair and/or Course instructor) to ensure that appropriate academic accommodations to program requirements are arranged. These accommodations may include individual counselling, alternative formatted literature, accessible campus transportation, learning strategy instruction, writing exams and assistive technology instruction.

Accommodation Policies

Students with disabilities are encouraged to contact Accessible Education, which provides recommendations for accommodation based on medical documentation or psychological and cognitive testing. The policy on Academic Accommodation for Students with Disabilities can be found at:

https://www.uwo.ca/univsec/pdf/academic_policies/appeals/Academic Accommodation_disabilities.pdf.